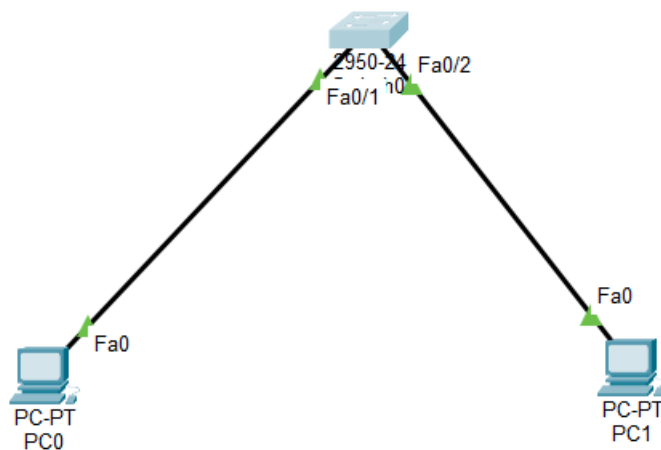


**Switching Basics Theory: Switch operation: MAC address table, frame forwarding.**  
**Practical: Observe MAC table population in a simulated network. Task: Explain the process of frame forwarding in switches.**

### Task 1.

Open Cisco Packet Tracer or any network simulator, create a simple network with one switch and two PCs, and use the 'show mac address-table' command to observe how the MAC address table is initially empty.



```
Switch#  
Switch#  
Switch#show mac address-table  
          Mac Address Table  
-----  
Vlan    Mac Address      Type      Ports  
----    -  
Switch#
```

## Task 2.

Send a ping from PC1 to PC2 in your simulated network, then immediately run 'show mac address-table' on the switch and note how the MAC addresses of both PCs appear in the table.<br><br><em><strong>Hint:</strong> Try pinging in both directions to see how the table updates.</em>

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.29.20

Pinging 192.168.29.20 with 32 bytes of data:

Reply from 192.168.29.20: bytes=32 time<1ms TTL=128
Reply from 192.168.29.20: bytes=32 time<1ms TTL=128
Reply from 192.168.29.20: bytes=32 time<1ms TTL=128
Reply from 192.168.29.20: bytes=32 time<1ms TTL=128

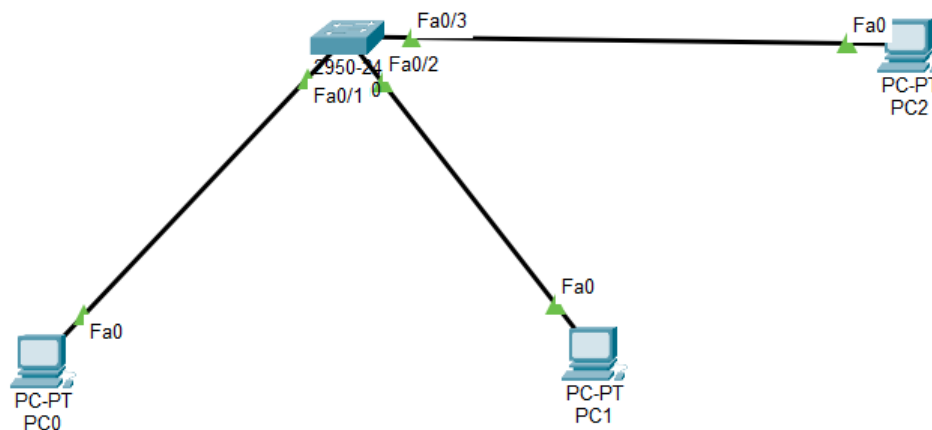
Ping statistics for 192.168.29.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

```
Switch#show mac address-table
          Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0003.e481.cd3d   DYNAMIC Fa0/2
1       0050.0f3c.acd1   DYNAMIC Fa0/1
Switch#
```

### Task 3.

Add a third PC to your network, connect it to the switch, and ping from PC3 to PC1. Observe and record how the MAC address table changes after this communication.



```
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.1756.9223    DYNAMIC Fa0/3
1       0050.0f3c.acd1    DYNAMIC Fa0/1
Switch#
```

### Task 4.

Suppose you see the following MAC address table on a switch: Port 1 - 00:11:22:33:44:55, Port 2 - 66:77:88:99:AA:BB. If a frame arrives at the switch on Port 1 with destination MAC 66:77:88:99:AA:BB, explain step-by-step how the switch will forward this frame.

| Port   | MAC Address       |
|--------|-------------------|
| Port 1 | 00:11:22:33:44:55 |
| Port 2 | 66:77:88:99:AA:BB |

**A frame arrives on Port 1 with destination MAC:**

**66:77:88:99:AA:BB**

**Step 1:**

The switch receives the frame on Port 1.

**Step 2:**

It examines the source MAC address and can learn/update which port that source MAC is associated with.

**Step 3:**

It examines the destination MAC address:

66:77:88:99:AA:BB

**Step 4:**

The switch searches its MAC address table.

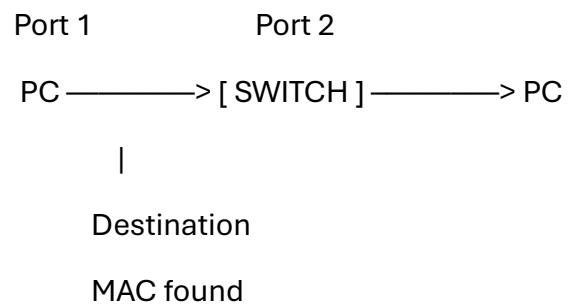
**Step 5:**

It finds:

66:77:88:99:AA:BB → Port 2

**Step 6:**

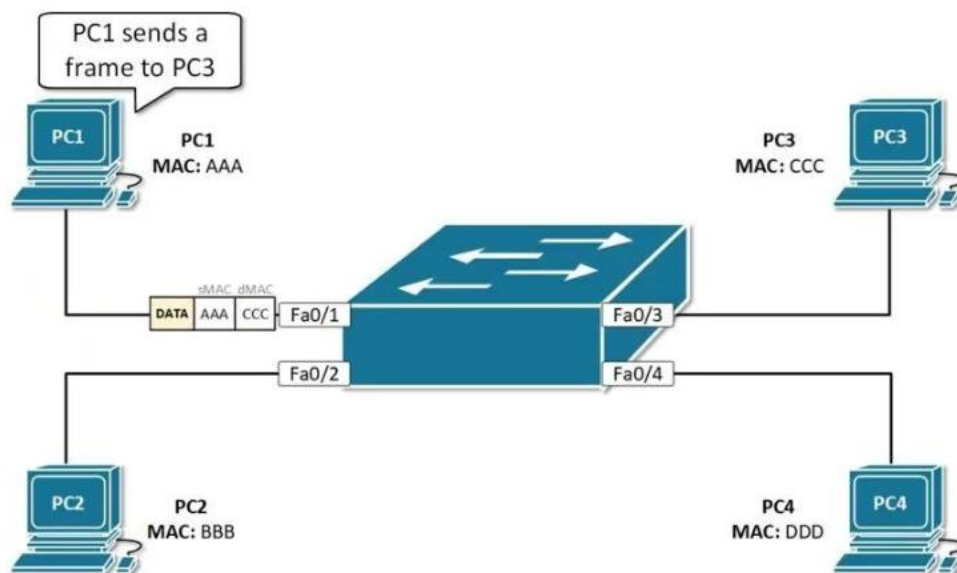
Because the destination is known, the switch forwards the frame only through Port 2.



The switch does not broadcast the frame to all ports because it already knows that the destination MAC is on Port 2.

### Task 5.

Draw a diagram showing how a switch learns and uses MAC addresses to forward frames in a network with 4 devices, and write a short explanation (3-4 sentences) describing the process in your own words. **Constraint:** Do not copy definitions from the textbook; use your own understanding and examples.



## Learning Mac addresses

| Device | MAC Address | Switch Port |
|--------|-------------|-------------|
| PC1    | AAAA        | Fa0/1       |
| PC2    | BBBB        | Fa0/2       |
| PC3    | CCCC        | Fa0/3       |
| PC4    | DDDD        | Fa0/4       |

- The switch learns a device's MAC address when it receives a frame from that device.
- It records the MAC address together with the port where the frame arrived.
- When another device sends a frame, the switch checks the destination MAC in its table and sends the frame through the correct port.
- For example, if PC1 sends data to PC3, the switch looks up PC3's MAC address and forwards the frame only to **Fa0/3**.